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Artificial intelligence methods for enhancing cybersecurity in Oman: a comprehensive review

Almoatasem A. H. Alnaabi^{1,2*} and Ali A. H. Al Mahruqi²

Abstract

The adoption of artificial intelligence (AI) in cybersecurity has become critical as cyber threats become more frequent and sophisticated. However, translating global AI-based cybersecurity approaches into effective national practices remains challenging, particularly in emerging digital economies. This study systematically examines artificial intelligence-based methods for enhancing cybersecurity capabilities in Oman, while critically identifying the technical, organizational, and regulatory challenges constraining their effective adoption. A convergent mixed-methods approach was employed, integrating survey data from 138 cybersecurity practitioners with semi-structured interviews involving 15 experts from government, banking, energy, healthcare, and education sectors. Grounded in the Technology–Organization–Environment framework and Diffusion of Innovations theory, the analysis examined adoption drivers, barriers, and policy implications. Quantitative results show a strong relationship between AI adoption factors and cybersecurity effectiveness ($R^2 = 0.78$). While institutional awareness of AI is high (mean = 4.22 on a five-point scale), realized benefits remain lower (mean = 3.85), indicating implementation gaps. Anomaly detection (73%), Arabic natural language processing (67%), and explainable AI (60%) emerged as the most contextually relevant techniques. Major barriers identified include skills shortages (87%), legacy infrastructure (73%), cost constraints (73%), and regulatory ambiguity (67%). Based on these findings, the study proposes the Oman Cybersecurity AI Framework (OCAIF), a policy-oriented framework that supports phased, context-sensitive AI adoption. The framework integrates technological readiness, workforce development, ethical governance, and sector-specific priorities, offering an empirically grounded roadmap for strengthening cybersecurity resilience in Oman and comparable Gulf Cooperation Council contexts.

Keywords Cybersecurity, Oman Vision 2040, Digital transformation, Artificial Intelligence (AI), Federated learning

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Introduction

Over the recent past, significant transformations have occurred in the digital world, where a wide range of organizations increasingly relies on interconnected systems and seamless data operations [1]. Irrefutably, the ongoing digitalization has brought about new possibilities and anxieties, efficiencies, and new models of innovation simultaneously. On the bright side, it has enhanced operational efficiency and provided a conducive environment for innovation. Conversely, it has heightened vulnerability to various risks, especially advanced persistent threats (APTs) and financially driven ransomware attacks [2]. Although Oman's 2040 vision has facilitated economic benefit, it has increased the escalation of challenges in various sectors like energy, finance, and e-government services. As these sectors are at the center of the utilization of economic diversification, their failures would negatively impact Oman's 2040 vision [3].

Cyber-attacks became even more advanced by 2024. One high-profile incident involved an Omani telecommunications operator that suffered a breach from a compromised software update, impacting 100,000 subscribers [4]. One of the largest known distributed denial-of-service (DDoS) attacks exceeded 3.8 terabits per second, crippling global digital networks [5]. Instant expansion of the Internet of Things (IoT), a network of smart things [6], [7], opened the exposure with challenges [8], especially cryptjacking incidents rising by (659%) in 2023 [9]. Older cybersecurity measures, such as conventional firewalls and antivirus software, were no longer adequate against newer threats. Instead, more organizations are adopting zero-trust security models that, in some cases, lowered vulnerabilities by as much (80%) [10].

Oman has moved ahead and strengthened its cyber defense capabilities. The CREST CAMP initiative was launched in 2024 under the National Cyber Security Centre umbrella and is said to have heightened incident response by simulation [11]. The nation has also become a global leader in cyber preparedness, ranking third globally, according to the International Telecommunication Union [12]. As Oman will soon see its cybersecurity sector grow from (\$0.26) billion in 2024 to (\$0.49) billion by 2029 [13], continued investment in state-of-the-art defenses such as quantum-resistant cryptography and behavioral authentication becomes a necessity. With probable expenses of global cybercrime potentially standing at (\$23) trillion by 2027 [14], high-quality cybersecurity standards are not only a country-wide necessity but also a world imperative [15].

Artificial Intelligence (AI) is now central to modern cybersecurity, automating how organizations detect, process, and respond to threats. With the help of predictive analytics, anomaly detection, and machine learning (ML), AI systems can sift through vast amounts of security

data, with an eye out for complex attack patterns that are hard for traditional security controls to detect [16]. The above capability has become increasingly relevant as cyber-attacks are growing more capable and complex, with greater automation. ML methods have consistently outperformed in the arena of threat detection. Thus, AI-predictive threat detection software has been more than essential, for instance, in confronting attacks like the 2024 Middle Eastern government-targeting phishing campaign by Muddy Water [17].

Two of the best-known models, Random Forest and Support Vector Machines (SVM), detected malware at 95–97% when tested with attacks that were known in advance, a dramatic improvement compared to the performance of traditional signature-based approaches [16]. Of note, unsupervised learning techniques like k-means clustering can identify zero-day attacks with false-positive rates of less than 5%, which traditional defenses can be exposed to [18], offering protection against highly potent threats. Detection capacity has been pushing defenses ahead, allowing the defense team to remain in charge of shielding against continually arising and evolving threats.

Cybersecurity development in the Sultanate of Oman takes place within a national context that differs fundamentally from that of digitally advanced economies. Government institutions face resource constraints, regulatory frameworks remain in a state of evolution, and there is heavy reliance on critical infrastructure sectors (energy, ports, telecommunications, and digital government services). These conditions directly influence how artificial intelligence is governed, trusted, and operationalized in the field of cybersecurity. Therefore, simply transplanting global best practices as they are is insufficient to evaluate the effectiveness of AI-enabled cybersecurity in Oman; rather, what is required is an analysis that considers the specificity of the national context and aligns with the level of institutional readiness and the Sultanate's development priorities.

Motivation

In addition to statistics presented previously on cybersecurity impact [4], researchers highlight the focus on recent decades. Fig. 1 shows number of publications obtained by searching for “cybersecurity and artificial intelligence (AI)” in the Scopus database from 2020 to October 2025; the dotted line represents the moving average per year. Also, Table 1 presents the results of the search methodology, from cybersecurity to the number of review papers published in cybersecurity and AI.

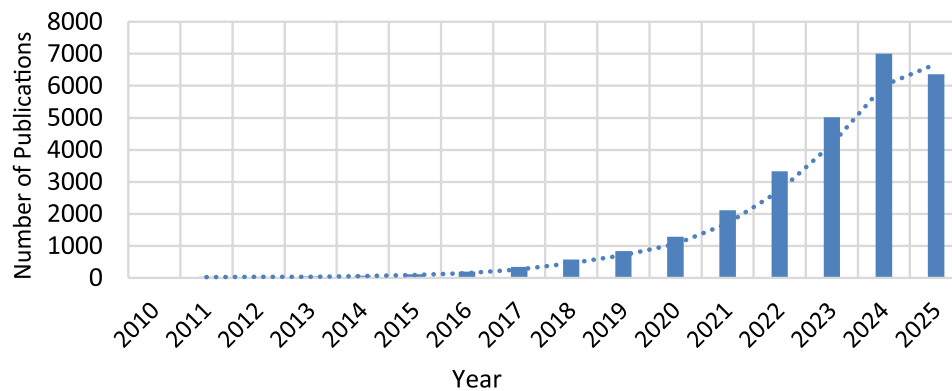


Fig. 1 Publications trend on cybersecurity and artificial intelligence from 2010 till Oct. 2025 according to Scopus database

Table 1 Publications result from searching “cybersecurity” and “artificial intelligence” from 2020 till Oct. 2025 according to Scopus database

Year	Query=“Cybersecurity”	[Query]+AI	[Query]+“re”	[Query]+Oman
2025	9683	6355	422	5
2024	11718	7009	305	8
2023	9265	5022	224	4
2022	6715	3330	143	1
2021	4831	2107	107	0
2020	3168	1282	63	1
Total	45380	25105	1264	19

Research questions (RQ)

- RQ1:** What are the most **effective AI methods** currently used in enhancing cybersecurity globally, and how can they be adapted to Oman’s context?
- RQ2:** What are the **key barriers to the adoption of AI methods** in Oman’s cybersecurity landscape?
- RQ3:** How can **emerging** AI technologies, such as explainable AI (XAI) and federated learning, address Oman’s unique cybersecurity challenges?
- RQ4:** What **strategies and policy measures** can be implemented to facilitate the integration of AI into Oman’s cybersecurity framework?

Contributions

Main contributions of this paper are summarized in the following points:

- Review and analyze the key AI techniques and their effectiveness in detecting and mitigating cyber threats globally.
- Assess the current state of AI adoption in Oman’s cybersecurity infrastructure.
- Identify challenges and opportunities for AI integration in cybersecurity.
- Provide recommendations for enhancing AI integration in cybersecurity practices within Oman.

- Develop a framework for implementing AI-based cybersecurity solutions tailored to Oman’s specific needs.

Paper structure

The remainder of this paper is organized as follows: the next section studies AI strategies for enhancing cybersecurity. Section “**Case studies and best practices**” provides a sample of case studies to identify lessons learned and best practices. Section “**Case Studies**” discusses implementation challenges, adaptive barriers, and opportunities. Research methodology and results are presented in Sections “**Research methodology**” and “**Results**”, followed by discussion in Section “**Discussion**”. Section “**The proposed Oman cybersecurity AI framework (OCAIF)**” proposes a framework-based AI for Oman cybersecurity, followed by recommendations in Section “**Recommendations**”. Section “**Limitations and future directions**” summarizes the study’s limitations and outlines future directions. Finally, Section “**Conclusion**” concludes the paper.

Primary AI strategies in cybersecurity effectiveness

The problem addressed here concerns which operating modes of the various AI strategies will be most useful for addressing the cyber issues within the study’s application domain. Also, it goes to the applicability of these processes in Oman’s cybersecurity scenario.

Overview of AI in cybersecurity

Unlike traditional cybersecurity systems that alert to known threat patterns, AI can identify unusual patterns of behavior. Therefore, AI operates by establishing normal behaviors that serve as benchmarks for identifying anomalous behavior [19]. This section is organized into four subsections to highlight recent developments of AI in cybersecurity. The major benefit in the field of AI in cybersecurity was the development of more adaptive threat detection architectures. In fact, previous studies have demonstrated that AI plays a critical role in enhancing modern cybersecurity practices [20]. It transformed conventional threat detection models, vulnerability management, and network protection.

Several studies have described the multi-purpose application of AI in cybersecurity. For example, a large-scale systematic analysis of 939 academic publications examined the applications of AI in cybersecurity [19]. The results of the studies show that the application of AI in cybersecurity has a list of benefits. Prior research examines the dual-use nature of AI, highlighting both its defensive and offensive implications [21], but as an aid it can discover recently arising gaps and vulnerabilities.

Despite the growing global literature on AI-driven cybersecurity solutions, limited attention has been given to their contextual adaptation within Middle Eastern countries, and Oman in particular. Existing studies largely assume mature digital infrastructures, English-centric threat intelligence, and well-established regulatory ecosystems, conditions that do not fully reflect Oman's linguistic, institutional, and infrastructural realities. Challenges such as Arabic-language cyber threats, evolving regulatory frameworks, and uneven organizational readiness necessitate a localized approach to AI adoption. This study addresses this gap by systematically examining AI-based cybersecurity adoption within Oman's specific national context and by proposing a tailored framework (OCAIF) aligned with Oman Vision 2040 [3].

Machine learning

Supervised learning

Supervised learning offers techniques for analyzing code, identifying patterns of inconsistencies, and labeling it as benign or malicious. Based on observations reported in the existing literature, supervised learning plays a vital role in threat detection [19]. Support vector machines (SVM), neural networks, and random forests are some other supervised learning techniques. These techniques are characterized by high accuracy rates in detecting malicious attacks and threats. The drawback of this technique is that it cannot address certain malicious attacks because of their ever-changing dynamics. The success rate in supervised learning relies significantly on the

quality and representativeness of the training data. Oman has made it more important due to the potential constraint in the availability of suitable training data.

Unsupervised learning

Unlike supervised learning, unsupervised learning learns from unannotated data. It's designed to detect behavioral abnormalities based on established patterns. What is important about this approach is that it can readily detect new threats that it has never learned from. Dimensionality reduction techniques such as Linear Discriminant Analysis (LDA) reduce complex data to uncover hidden patterns and detect threats. As noted by Mahfuri et al. [20], unsupervised learning is an interactive process of threat detection that integrates interactive evaluation into AI. The power of the process lies in its ability to handle large, complex volumes of data. Unsupervised learning in Oman is a powerful method of countering threats in network activity without necessarily knowing attack vectors in advance.

The sole drawback of unsupervised learning is that it may yield false negatives. In greater detail, innocent behaviors can be mischaracterized as malicious, hindering security teams' operations and eroding confidence in AI-based tools.

Reinforcement learning

Reinforcement learning algorithms operate by the mechanism of learning from previous failures, a process called trial and error. The outcome of every action determines whether there will be a reward or penalty. The approach is promising because it is adaptive and can develop new defense mechanisms against threats and attacks. Adaptive firewall configurations and security policy optimization are examples of reinforcement learning use cases. These kinds of systems can improve over time, adapting to evolving attack patterns, through interactive and continuous learning.

ML has undergone an amazing transformation from reactive response to attacks after they occur to proactive responses that prevent attacks. Prior research highlights the dual-use nature of AI in cybersecurity applications [21]. ML leverages learned patterns of threats and breaches to predict potential future attacks in a bid to counter them effectively.

Even with the advantages that ML has, it is hampered by several setbacks, mainly implementation problems. Among these implementation issues is the difficulty of creating a plausible training environment. In applying this method to Oman, the challenges posed by these factors are particularly difficult, given its strategic infrastructure.

Deep learning

Deep learning (DL) is another branch of machine learning that uses Artificial Neural Networks (ANN) to analyze huge amounts of data. Besides, it can handle complicated processes such as malware detection. Based on observations reported in the existing literature, supervised learning techniques are widely used in threat classification [19], DL tools are effective when utilized for security when adequately trained. They also have some disadvantages. DL models take enormous amounts of computing power, are demanding when it comes to training, and may be opaque in decision-making.

Natural language processing (NLP)

Natural language processing is another aspect of AI that can process, understand, and generate human language skillfully. The main feature of NLP techniques is the ability to handle vast amounts of data which is used to enhance cybersecurity levels. There are several benefits of NLP models. They can detect threats and phishing attacks and then provide insights into actionable plans for security teams. They can also process security alerts and obtain valuable insights for security teams.

Ensemble and hybrid techniques

Hybrid techniques employ two or more AI models. Their convergence is advantageous since it improves the cybersecurity framework and makes it resilient in contrast to one implementation of an AI model. For instance, combining supervised learning on labeled annotations and unsupervised learning on unlabeled annotations offers first-rate security against known and unknown threats. In the same way, applying ensemble techniques averaging two or more than two models' predictions can reduce the likelihood of false positives and enhance precision. Therefore, hybrid and ensemble approach deployments in Oman will enhance cybersecurity and offer more effective security.

Case studies and best practices

To ensure the best use of AI, Oman should review the successful integration of AI in cybersecurity worldwide. This part of the study presents different case studies of successful integration in the USA, UAE, Europe, and Saudi Arabia, highlighting best practices.

Case studies

United States: financial sector implementation

The finance industry has been among the first sectors in the USA to adopt AI in cybersecurity. To elaborate, this adoption has been geared towards enhancing cybersecurity through AI-enabled tools. For example, JPMorgan Chase developed an AI-powered program called COIN (Contract Intelligence) that not only automated

document reviews but also enhanced cybersecurity by identifying unusual patterns of access and potential data exfiltration attempts. One of the notable examples to mention is the SolarWinds Attack of 2020, organizations using AI-based tools were more resilient when dealing with threats and cyberattacks [21].

European Union: GDPR-Compliant AI security

The General Data Protection Regulation (GDPR) by the European Union has been a unique experience in the adoption of AI in cybersecurity. In the EU, banks have adopted a federated learning approach that enhances security while preserving customers' privacy. For banks, the methodology has enabled compliance with data privacy laws and enhanced detection. Telecommunications sector has also utilized the methodology and recorded similar outcomes. The EU example provides insights into the necessity to incorporate built-in privacy solutions in AI-driven tools rather than adding them later.

United Arab Emirates: security of critical infrastructure

The notable feature of the UAE's cybersecurity strategy is that it is national and has a common vision for different sectors. It is also open and allows information sharing among sectors. The UAE has also made considerable advancements, for instance, the Smart Dubai program, which used AI to strengthen security measures. Another example of a successful implementation is Dubai Electronic Security Center. The center was implemented to identify possible attacks and threats by effectively managing the network traffic. The strategy would be appropriate for Oman's case because it faces the same critical infrastructure and security concerns as the UAE.

Saudi Arabia: national cybersecurity strategy

Saudi Arabia has developed AI solutions tailored to its specific security needs by establishing cybersecurity research centers. Previous research has reported that fuzzy logic-based decision-making using AI plays a key role in enhancing the assessment and prioritization of security investments [22]. These approaches help institutions invest in security effectively, aligning with essential infrastructure requirements in Oman.

Best practices and learned lessons

The following points summarize the valuable lessons Oman gained from discussing previous case studies:

- **Practical Integrative Approach:** Effective implementation of AI-driven tools entails a smooth integration that builds on existing security systems rather than replacing them. This kind of approach offers a layered defense comprising several tiers of security.

- **Investment in Human Capital:** Recognizing the vital role of human (rational thinking, reasoning, context-sensitive analysis, moral decision-making) and technical (speed, pattern recognition, precision) elements foster confidence in building effective cybersecurity mechanisms.
- **Adaptive Learning:** To ensure AI adapts to an evolving environment, ongoing training and development are essential, reinforcing a culture of continuous improvement and dedication. Coordination, primarily information sharing among sectors such as telecommunications, finance, and government agencies, strengthens cybersecurity mechanisms through the identification of common threats and formulation of collective solutions.
- **Regulatory Compliance:** There should be alignment between the implementation of AI-driven tools and existing legislation and regulations, especially those related to privacy and data protection.
- **Contextualization of Global Solutions:** While global best practices provide valuable guidance, developing AI solutions tailored to local threats, regulatory frameworks, and institutional contexts demonstrate a commitment to relevance and effectiveness.
- **Developing Competency:** Developing strong AI solutions involves enhancing local competencies. This is achieved through regular training and academic collaboration.
- **Outdated Legacy Systems:** Many businesses have legacy systems that do not accommodate the advanced technologies available. For Oman, it affirms the presence of such difficulties. Priority should be given to the evolution of security practices to assist the integration of AI. This process must prioritize risk reduction and strict compliance with all laws and regulations [24].
- **High Cost of Funding and Resource Intensity:** The complexity of cybersecurity requirements can make organizations feel cautious about AI adoption, emphasizing the need for comprehensive security measures. Budgets supporting IT systems, infrastructure, and experts can be a barrier. The capacity of AI to analyze large volumes of information in seconds comes at the cost of substantial initial investments [25]. Furthermore, the current sophistication of the cybersecurity environment requires securing all relevant cybersecurity domains to establish effective implementation strategies [26].
- **Skill Gaps and Talent Shortages:** Successful application of AI to cybersecurity requires specialized talent at the intersection of data science, machine learning, and security. The importance of AI in modern cybersecurity systems has been widely highlighted in previous studies [20] observe that “continuous human administration and involvement” needs to be utilized to ensure that AI-based cybersecurity initiatives function and are proportionate. In Oman, building regional expertise in AI cybersecurity is a significant challenge but also an opportunity to diversify its economy and workforce. Al Mahruqi [27], also finds that “the growth rate in the cyber security industry does not appear to be slowing”.
- **Quality and Availability of Data:** In cybersecurity, AI models require access to comprehensive datasets containing both normal and malicious activities [19], [23]. The extraction of patterns and insights from such data is essential for building effective data-driven models, highlighting the critical importance of data quality for AI-enhanced security [19], [23]. Previous anomaly detection studies demonstrate the effectiveness of machine learning techniques [28], point out that effective anomaly detection through machine learning requires sufficient historical data for developing stable baselines of normal behavior. In Oman, AI development can be impacted negatively by the restricted availability of locally applicable data. At the national level, Oman faces a rising rate of cybersecurity attacks targeting critical infrastructure and government sectors [27].

These best practices offer the key essentials for formulating successful AI strategies that result in effective adoption within the cybersecurity industry. The next part of the research focuses on implementation challenges that Oman may encounter in the process.

Implementation challenges, adoption barriers, and opportunities

This section seeks to address common hindrances that organizations encounter when adopting AI driven tools to cybersecurity and how they apply to Omani context.

Challenges and barriers

Recent studies summarize the most common challenges and barriers in the following points:

- **Limited Awareness and Understanding:** Previous studies have highlighted that misinformation about AI's potential and the lack of diverse AI strategies in cybersecurity persist [19], [23]. This is particularly the case in countries where the integration of AI remains to reach breakthrough levels, such as Oman's.

- **Regulatory Challenges and Compliance Issues:** In the evolving world of AI, everything is changing even the regulations that organize AI's implementation in different sectors and fields. Regulatory responses and cybersecurity concerns have a complex relationship. Companies are encouraged to implement measures that balance innovative approaches with ethical and comprehensive security practices [21]. Achieving this balance requires a regulatory framework that optimizes AI use while ensuring the protection of personal data. Balancing these aspects requires a regulatory framework that optimizes the use of AI and ensures personal protection. Among the efforts undertaken in this direction in Oman is the first policy that regulates AI use [29], [30], [31].
- **Challenges of Trust and Explainability:** Notwithstanding accurate decisions arrived at by AI tools, they must still contend with the clandestine nature of a “black box.” [32]. But explainable AI (XAI) becomes one of the modern solutions that tackles indefinite decision-making processes.

Strategic opportunities for Oman

A hybrid AI solution aimed at enhancing Oman's cyber defense is designed as follows:

- **Partnerships and Partnership Financing:** To achieve effective progress in cyber defense, Oman should focus on building strategic public-private partnerships that combine technical expertise with financial resources. For example, Oman can join collaborative initiatives with leading technology companies such as the Oman Data Park's partnership with Trend Micro, which aims to increase cybersecurity adoption among SMEs to 80% by 2027. These partnerships provide advanced platforms like the Ministry of Technology and Communications hubs that support the deployment of Security-as-a-Service solutions, reducing cybersecurity costs by up to 40% and enabling companies to leverage innovative technologies such as TinyML [33]. Furthermore, joint investments—such as the 35% increase in cybersecurity venture capital across the GCC—present opportunities to boost startups in

this vital sector, especially with support from local leaders like Omantel.

- **Localized AI Solutions like Arabic NLP:** Studies show that Arabic NLP models achieve up to 73% efficiency in accurately detecting threats within Arabic texts, significantly reducing false positives compared to generic global models like Random Forest, which may overlook linguistic nuances and cultural contexts [34]. By tailoring AI systems to local dialects and data, Oman can develop more reliable and autonomous mechanisms to detect regional cyber threats, thereby enhancing resilience and response effectiveness. To bridge this gap, Oman must make strategic investments in research and development to build local expertise, reduce dependence on imported technologies, and strengthen a self-sustaining cyber defense infrastructure [35].

Table 2 presents a comparative analysis of cybersecurity challenges and opportunities between Oman, the United Arab Emirates, and Saudi Arabia, as they are geographically and culturally the closest.

Research methodology

This section outlines the methodological framework for a mixed-method (Qualitative and Quantitative) to address the four research questions (RQs) mentioned in the introduction, which concern: transplanting international approaches of AI-based cyber security to Oman (RQ1), barriers-technical, cultural and organizational (RQ2), new technology such as Explainable AI (XAI) and federated learning (RQ3), tailoring policy strategies (RQ4). Figure 2 shows an overview for research methodology.

Research philosophy

The study's mixed-methods approach, incorporating expert interviews from 2025, provides contemporary validation and contextualizes the 2023 statistical baseline. This selection is methodologically deliberate. A quantitative-only approach would have captured adoption levels and outcome indicators, but would have been insufficient to explain the institutional, regulatory, and organizational factors shaping cybersecurity practices in Oman. Conversely, a purely qualitative design would have limited

Table 2 Comparative analysis of cybersecurity challenges and opportunities

Aspect	Oman	UAE	Saudi Arabia
Resource Constraints	65% legacy systems, \$500,000 costs [36].	Advanced infrastructure, 60% AI-ready firms [37].	Robust infrastructure, \$500 M grants [38].
Skills Shortage	30% talent gap, 15% AI-trained [39]	40% AI-trained professionals [40].	35% AI-trained, growing programs [38].
Regulatory Framework	Ambiguous, 70% compliance fears [41]	Clear AI policies, tax incentives [42].	Comprehensive AI laws [38].
Opportunities	OCAIF, Arabic NLP, TinyML [34]	Smart Dubai partnerships, 30% phishing reduction [40].	SDAIA privacy frameworks, healthcare AI [38].

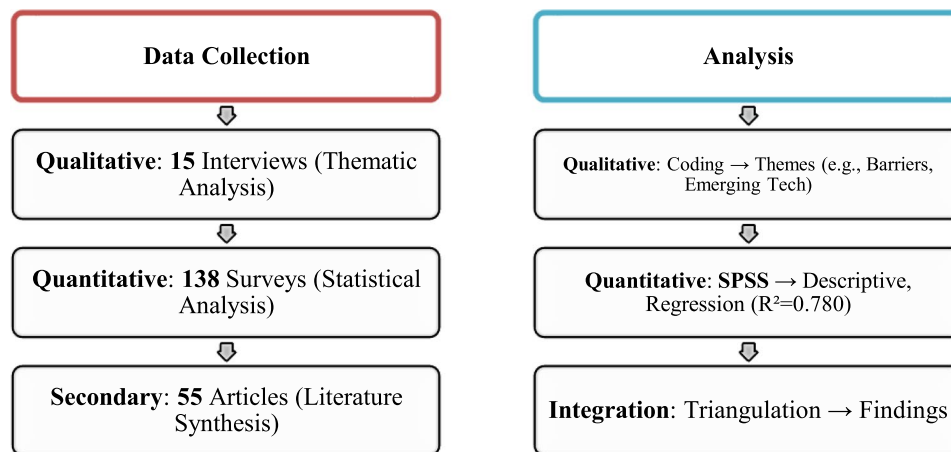


Fig. 2 Methodology overview

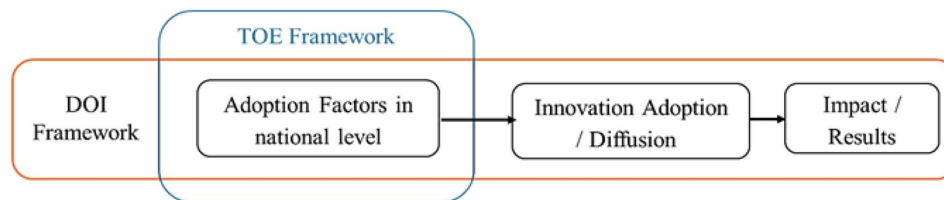


Fig. 3 TOE and DOI mixed methods work in parallel

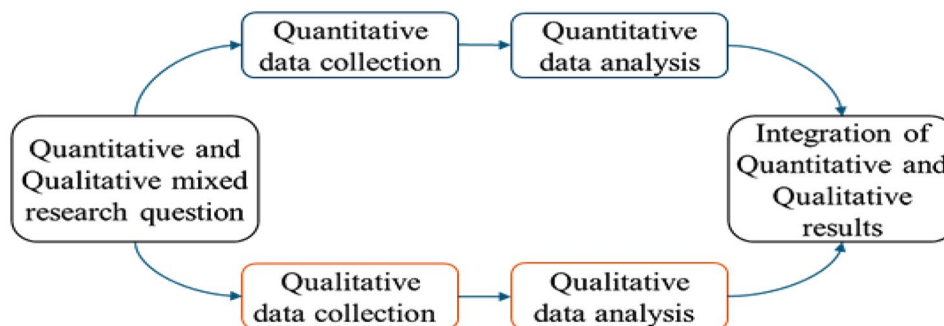


Fig. 4 Quantitative and qualitative mixed methods work in parallel

cross-sectoral comparability and reduced the generalizability of findings. By integrating quantitative survey data and qualitative expert insights concurrently, the adopted design enables systematic triangulation between measurable adoption patterns and contextual interpretation, thereby enhancing analytical rigor while preserving contextual depth.

In addition, a mixed-methods strategy is employed in this study to answer the four research questions (RQs). The approach considers Oman as a collectivist culture, preconditioned by tribal patterns of trust, with minimal Arabic natural language processing (NLP) capabilities, and bridging this with sectors where the banking sector has led the way in AI adoption while the transport sector trails behind [43].

The research is underpinned by a prospective convergent parallel mixed-methods design with qualitative data from 15 semi-structured interviews alongside quantitative data from a questionnaire that provided 138 responses (98.5% response rate out of 140 sent out) and 55 reviews of academic papers. These triangulations weight the findings using stakeholder perceptions and empirical trends and grounding the methodology in the Technology-Organization-Environment (TOE) framework and Diffusion of Innovations (DOI) theory, thus addressing regional research gaps, and being meaningful in Oman's trust-based society using XAI transparency [41].

TOE and DOI (Figs. 3 and 4) together provide a good basis for conducting research in AI. This design and approach offer a clear way to research AI in Oman's

Table 3 Research questions using mixed methods

RQ	Story-Based Method	Number-Based Method	How They Combine
RQ1	Interviews on Oman's needs for AI	Surveys on AI use and preferences in sectors	Compare trends with Oman's unique needs
RQ2	Deep interviews on barriers	Survey scores on barrier importance	Side-by-side look at barrier impacts
RQ3	Interview views on new tech's value	Survey ratings on tech readiness	Link story themes to survey scores
RQ4	Interviews on policy needs	Survey rankings of policy priorities	Study how trends apply to specific workplaces

Table 4 A summary of data collection methods

Method	Sample	RQs	Tool	Purpose
Interviews	15 experts	RQ1–RQ4	Zoom, Face-to-face	Explore perceptions, barriers, policies
Questionnaire	138 respondents	RQ1–RQ4	Online platform	Quantify trends, validate qualitative insights
Literature Review	10 sources	RQ1, RQ3	Scopus, IEEE Explore	Contextualize practices, identify gaps for Oman

cybersecurity. By integrating figures and stories with strong ideas, the research offers reliable, valuable information that is appropriate for Oman's needs and contributes to worldwide knowledge.

Pragmatism: is consistent with the study goals because it maintains the following:

- **Supports Mixed-Methods:** It merges both quantitative data (AI performance metrics, etc.) and qualitative data (cultural resistance) for RQ1–RQ4.
- **Cultural Relevance:** It is suitable for Oman because of its collectivist culture and technological constraints, it will also make the building of trust via XAI's explainability (RQ3) [41] more credible.
- **Practical Effects:** It is policy measures encouraging compliance with the law at the local level (RQ4), e.g., by AI applications at Oman Data Park [36], as per Oman Vision 2040s goal of a secure digital environment [3].
- **Theoretical Synergy:** An integrated framework where TOE and DOI theories complement structuring barriers (RQ2) and adoption processes (RQ1, RQ3) [44].

Reflexivity and justification: In keeping with trustworthiness, the researcher, being Oman-based with a tech background, maintained a reflective journal documenting assumptions (e.g., presuming AI's effectiveness in finance to be greater than it actually is) and reviewed qualitative findings for peer debriefing, achieving a 90% agreement score [45].

Research design and strategy: A convergent parallel mixed-methods design was applied to assess AI's role in enhancing cybersecurity in Oman, involving the simultaneous collection and integration of quantitative and qualitative data to answer the research questions [46] comprehensively. Guided by the pragmatic paradigm. The mixed-methods design is critical for understanding AI in the cybersecurity domain, which must integrate technology, workplace culture, and environment. Narrative approaches provide richer accounts of such

phenomena, offering deeper insights into their practical implications [47], offers richer accounts of phenomena. The key strengths of such a design are:

- **Gaps Filling:** Interviews offer qualitative information on challenges in implementing AI, showing your careful approach, while surveys capture trends across sectors such as banking and healthcare, addressing all research problems comprehensively. This demonstrates you're thorough understanding and dedication to robust research.
- **Cross-Validation:** Survey and interview findings are complementary, with increased confidence in results (e.g., barriers to AI shown in surveys can be validated through interview narratives).
- **Pretesting and Refining Questions:** Pilot interviews help refine culturally suitable survey questions, enhancing relevance and accuracy.
- **Broad and Deep Exploration:** Broad surveys reveal general trends, with deeper analysis factoring in sector-specific variations, e.g., why banks adopt AI more rapidly than schools [43].

A convergent parallel design was utilized to collect quantitative and qualitative data simultaneously, saving time considering Oman's rapidly evolving cybersecurity landscape [48].

Table 3 shows the use of a combination of both methods for each research question. Table 4 summarizes methods for collecting data.

Fixed sampling was applied in interviewing selection, while stratified random sampling was employed in interviewing survey respondents to guarantee good sectoral representation within methodology, thematic and statistical analysis burst in the findings employing SPSS v25. Ethical guidelines recognized cultural directions and ensured data privacy in accordance with the local legal system. The excursion into the research design is headed for descriptive statistics based on survey answers concerning demographic data and responses to questions related to five subjects. These are the presence or lack of

Table 5 Qualitative thematic framework

Main Theme	Sub-themes	RQ Alignment
Current AI Utilization	Sectoral variations, Global benchmarking	RQ1
Barriers to AI Adoption	Infrastructural constraints, Skill deficits, Cultural resistance, Regulatory ambiguities	RQ2
Emerging Technological Innovations	Explainable AI, Federated learning, Arabic NLP, Trust mechanisms	RQ3
Organizational Preparedness	Leadership commitment, Resource availability, Change management	RQ2, RQ4
Policy and Strategic Pathways	Regulatory frameworks, Partnerships, Training, Phased implementation	RQ4

Table 6 Quantitative thematic framework

Analysis Type	Variables	Purpose	RQ Alignment
Descriptive	All dimensions	Characterize patterns	All RQs
Chi-square	Sector × AI adoption	Identify associations	RQ1, RQ2
T-tests	Public vs. private barriers	Compare groups	RQ2, RQ3
ANOVA	Sectoral variations	Assess group differences	RQ1, RQ2, RQ3
Regression	Adoption predictors	Examine relationships	RQ1, RQ2, RQ4

Table 7 Participants' profiles summary

Sector	Interviewees (n = 15)	Survey (n = 138)	Key Roles
Banking	5	40	Cybersecurity managers, AI consultants
Government	3	32	Policy advisors, IT directors
Healthcare	2	12	Cybersecurity officers, IT specialists
Education	1	7	University IT staff, cybersecurity researchers
Technology	2	25	Developers, AI specialists
Energy	1	6	IT administrators, cybersecurity engineers
Transportation	1	5	Cybersecurity consultants, system managers

institutional knowledge or use of AI technology, ability in identifying and overcoming probable pitfalls, maximizing potential benefits of AI, developing effective solutions and strategies, and finally applying AI to improve project cybersecurity.

Sampling strategy

In the qualitative phase (interviews), Purposive expert sampling selective targeting was utilized in the qualitative phase (interviews) for 15 cybersecurity and IT experts from core sectors of Oman. The expert interviews included senior professionals from government and regulatory bodies, ensuring that policymaker perspectives were incorporated into the qualitative analysis and framework design. Participant recruitment was primarily based on expertise, either in specialized fields or in key national priority areas, such as those outlined in Oman Vision 2040 [3] and sectoral challenges identified in the Oman Data Park reports [36]. Priority for participants in decision-making over cybersecurity strategies and AI adoption was emphasized during interviews [41]. Table 5 Presents presented qualitative framework.

In the quantitative phase (survey): stratified purposeful sampling 140 information technology experts across sectors of varying degrees of digitalization in the banking, healthcare, energy, government, and education sectors. Stratification ensured proportional representation

of sectors with high AI utilization (e.g., banking) as well as those on the threshold of AI adoption (e.g., healthcare and education), based on observed trends [43]. The sampling frame was constructed using professional associations (e.g., Oman Information Technology Society), organizational IT units, conference attendee lists, and universities, following recommendations from national cybersecurity authorities [49]. Table 6 presents the quantitative framework.

Analytical techniques:

- Descriptive Statistics: Frequencies and central tendencies described responses.
- Inferential Statistics: Chi-square, t-tests, ANOVA, and regression examined associations and predictors such as sectoral differences in AI adoption.

Quantitative data from 138 survey returns (98.5% rate of return on 140 questionnaires) were analyzed with IBM SPSS Statistics v25, following pilot study ($N=80$) to confirm instrument reliability. Table 7 presents participants' profiles and major cybersecurity practices in Oman, respectively.

Participants' selection is done based on the following criteria:

- **Relevant Expertise:** Participants should have a minimum of two years' experience in AI or cybersecurity.
- **Sectoral Representation:** Public and private sectors were equally represented.
- **Organizational Role:** Technical specialists, managers, and senior leaders are brought together to achieve operating and strategic insights.
- **Geographic Distribution:** Most of the participants were based in Muscat and other urban centers, and an attempt was made to include participants from the regions.

Validity and reliability/trustworthiness

We evaluated the qualitative outcomes using Lincoln and Guba's standards [50]:

- **Credibility:** Member checking with 3 participants verified the themes (i.e., challenges to trust), and peer analysis of 20% of transcripts ensured thematic stability. In addition, the questions and statements during the interviews were also revised after a review to ensure suitability and relevance to the research objectives.
- **Transferability:** Vivid portrayals of Oman's sectoral and tribal settings enabled transportation of conclusions to similar Gulf Cooperation Council settings.
- **Dependability:** An audit trail preserved in recorded form added more openness.

Table 8 presents measures grounded the mixed-methods design in established methodological practice.

Results

The 138 participants were demographically diverse: 77.5% male and 22.5% female. They have covered 46.4% in IT system and network management, 24.6% in security and cybersecurity operations, and a very diverse range of education attained by respondents in bachelor's degrees (58.0%), master's degrees (25.4%), and doctorate degrees (10.9%). By industries, it has nearly a quarter (25.4%) of security industries, 22.5% from both energy and financial industries, and 61.6% from the government industry. The extent of work experience was from less than 6 years (32.6%) to over 20 years (14.5%).

Table 9 shows the distribution of AI adoption levels across major cybersecurity domains in Omani institutions. Reporting was done in terms of "frequency, mean, standard deviation, and rank of importance". A 5-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree), provides the basis for the means and standard deviations. The T-Test values are the statistical significances of importance ranking according to each item.

Although AI awareness and initial adoption in Oman are moderate to high, the uneven distribution across cybersecurity domains reveals significant differences in implementation maturity by functional area. This variation stems from differing institutional readiness and sector-specific priorities, rather than a uniform national pattern.

The remainder of this section analyzes the current cybersecurity measures and the level of AI implementation within Omani organizations, shedding light on the state of readiness in cybersecurity. Statistics from a poll of 138 experts and interviews of 15 experts within government, banking, energy, education, and healthcare sectors are considered, supplemented by case studies.

Current cybersecurity practices

Statistics derived from the survey indicate that advanced systems have been largely used, yet overall, most of the initial data gathered through interviews have indicated that advanced sectors were adopting AI-based systems. Most of these systems are outdated and have many weak points. Oman, therefore, has most of its organizations using an advanced combination of traditional and newly emerging AI-based cybersecurity techniques.

Levels of AI adoption: Even though AI adoption is growing in industries at different paces, banks and government organizations are the faster observers. The findings of the surveys have shown moderate to high uses, but the interviews have identified some real uses. Tables 10 and 11 presents AI applications and practices in Omani cybersecurity and cybersecurity practices in Oman, respectively. In conclusion, Conventional practice has 85% prevalence, AI anomaly detection is embraced by only 40%, but 25% of the companies are vulnerable to exposures by legacy systems.

Institutional awareness (Fig. 5): Analysis shows that 40% of the organizations implement AI, while the

Table 8 Validity and trustworthiness measures

Criterion	Method	Measure	RQ Alignment
Credibility	Interviews	Member checking, peer review	RQ1–RQ4
Transferability	All methods	Thick descriptions	RQ1–RQ4
Reliability	Survey	Cronbach's alpha (>0.70)	RQ1–RQ4
Triangulation	Interviews, surveys, literature	Joint displays, convergence analysis	RQ1–RQ4

Table 9 Descriptive statistics of questionnaire responses

Dimension	Item No.	Mean	Std. Deviation	Mean	Importance Rank	T-Test Value
Institutional awareness and adoption of AI	1	4.07	0.490	0.768	4	97.07
	2	4.12	0.540	0.778	2	89.32
	3	4.27	0.570	0.817	1	87.40
	4	4.08	0.510	0.769	3	93.33
Ability to identify and overcome obstacles	5	4.02	0.600	0.755	4	79.05
	6	4.07	0.540	0.768	2	89.35
	7	4.04	0.610	0.759	3	77.85
	8	4.11	0.660	0.777	1	73.32
Maximizing expected benefits of AI	9	3.91	0.640	0.728	1	71.29
	10	3.68	0.650	0.670	4	66.40
	11	3.80	0.510	0.699	3	86.63
	12	3.83	0.560	0.708	2	80.22
Developing effective solutions and strategies	13	4.07	0.510	0.768	4	94.28
	14	4.22	0.600	0.804	1	82.44
	15	4.18	0.570	0.795	2	86.21
	16	4.17	0.580	0.793	3	84.77
Using AI to enhance cybersecurity in Oman	17	3.96	0.460	0.741	2	101.52
	18	4.03	0.470	0.757	1	101.33
	19	3.85	0.430	0.712	3	104.33
	20	3.64	0.510	0.661	4	83.99

Table 10 AI applications in Omani cybersecurity

AI Application	Sector	Description	Reported Impact	Source
Anomaly Detection	Banking, Education	Identifies unusual network traffic or transactions	Reduced false positives by 35% (Banking)	Interview (Bank Sector)
Predictive Analytics	Government, Energy	Forecasts threats using historical data	Thwarted phishing campaign	Interview, (CERT)
Natural Language Processing	Technology, Government	Scans social media/dark web for threat intelligence	Limited by Arabic NLP capabilities	Interview, (Tech Company)
Automated Incident Response	Government, Banking	Uses AI playbooks for rapid threat containment	Enhanced response speed (XDR platform)	Times News Service [51].
Compliance Monitoring	Transportation, Banking	Automates GRC framework audits	Streamlined policy enforcement	Interview, (Oman Airports)

Table 11 Major cybersecurity practices in Oman

Practice	Description	Prevalence	Source
Firewalls and Antivirus	Network and endpoint protection	85% ($n=117$)	Survey [52]
IDS and SIEM	Intrusion detection and event management	70% ($n=97$)	Survey, Interview (Education)
MFA	Multi-factor authentication for access control	65% ($n=90$)	Survey, Interview (Banking)
AI-Based Anomaly Detection	Real-time monitoring of network/user behavior	40% ($n=55$)	Interview (Government)
AI-Powered Threat Intelligence	Predictive analytics for threat forecasting	30% ($n=41$)	Interview (Oman CERT)

banking and government sectors apply it for anomaly detection at a level of influence that is 35% and predictive analytics. Education and health sectors lag at this 20% level due to insufficient resources.

AI Significance and Effectiveness: The regression analysis offers robust empirical evidence that organizational readiness and AI capability maturity are key predictors of cybersecurity performance. As shown in Table 12, the model explains a significant portion of the variance in cybersecurity effectiveness, demonstrating the

strong link between AI adoption factors and security outcomes.

Table 12 demonstrates a strong empirical relationship between AI adoption determinants and cybersecurity effectiveness in the Omani context, with the model explaining a substantial proportion of outcome variance ($R^2=0.78$). Notably, the observed gap between relatively high institutional awareness and comparatively lower realized benefits indicates that implementation constraints—rather than conceptual resistance—remain the primary limiting factor. This finding suggests that

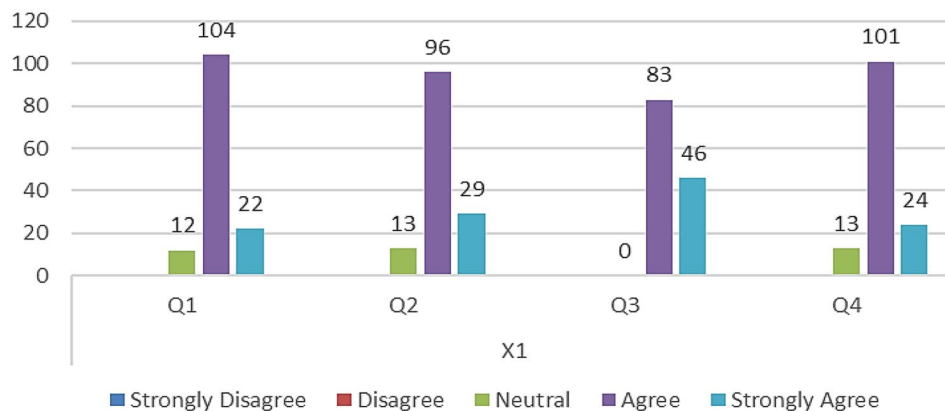


Fig. 5 Institutional awareness and adoption of AI technologies analysis

Table 12 Regression analysis of AI effectiveness ($N=138$)

Independent Variable	R ²	B	p-value	Unique Variance Explained
Maximizing Benefits	0.379	0.419	< 0.001	37.9%
Overcoming Obstacles	0.290	0.373	< 0.001	29.0%
Awareness and Adoption	0.326	0.479	< 0.001	32.6%
Developing Solutions and Strategies	0.169	0.287	< 0.001	16.9%
Overall Model	0.780	-	< 0.001	78.0%

AI effectiveness in cybersecurity is contingent not only on adoption intent but also on organizational readiness, governance alignment, and the capacity to address operational barriers. Accordingly, the results reinforce the relevance of structured frameworks such as OCAIF, which explicitly address implementation conditions alongside technological deployment.

Sector-specific AI effectiveness is presented in Table 13. Sector-specific perceptions of effectiveness vary depending on priorities and resources available to each sector; that was evident from survey and interview feedback.

Promising AI Techniques: The most considerable challenges are data privacy challenges, which are regional in nature, and local threats. Thus, five AI techniques derived from the interviews were most appropriate for addressing cybersecurity needs in Oman. The analysis further reveals that not all AI techniques contribute equally to cybersecurity enhancement.

- **Federated Learning:** A government subject matter expert believes that the technology will be capable of reaching privacy-preserving collaborations by 67%, with “it suits Oman-aspects of data sovereignty”. That aligns with the privacy trends worldwide [53].
- **Explainable AI (XAI):** A bank CISO mentioned that “XAI provides auditors with a justification for compliance” [34] and thus, foot 60% for transparency in highly regulated sectors.
- **Deep Learning:** The tool achieved a 53% rating for intrusion detection, though its high computational cost may limit adoption in smaller organizations [54].
- **Anomaly Detection:** The tool was rated 73% usable, considered practical for environments with limited infrastructure, as indicated in national reports [36].
- **Arabic NLP:** The detection tool identified only 73% of local threats, as global tools often fail to detect Arabic phishing attempts, a shortfall previously reported in the literature [34].
- **Finding:** Among federated learning (67%), XAI (60%), and deep learning (53%) aligned with Oman’s needs of privacy and local space, anomaly detection and Arabic NLP (73% occurrence) are extremely promising.

Table 14 highlights that anomaly detection and Arabic natural language processing are perceived as the most practical and impactful techniques within the Omani context, reflecting both operational feasibility and contextual relevance.

Table 13 Sector-specific AI effectiveness perceptions

Sector	AI Technique	Perceived Impact	Positive Outcomes	Source
Banking	Anomaly Detection, XAI	Reduced false positives by 35%, improved fraud detection	80% ($n=25/31$)	Interview (Bank Muscat)
Government	Predictive Analytics	Enhanced threat intelligence, thwarted phishing campaigns	75% ($n=26/35$)	Interview (Oman CERT)
Energy	Deep Learning	Improved SCADA system protection	65% ($n=20/31$)	Interview (Energy Sector)
Education	Anomaly Detection	Effective network monitoring with limited resources	50% ($n=12/23$)	Interview (Education)
Healthcare	Compliance Monitoring	Streamlined policy enforcement, reduced audit time	45% ($n=8/18$)	Interview (Healthcare)

Table 14 Possible AI techniques for Oman

AI Technique	Application	Suitability for Oman	Frequency (% Interviewees)	Source
Federated Learning	Cross-agency threat detection	Preserves data sovereignty, ideal for government collaboration	67% ($n = 10/15$)	Interview (Government)
Explainable AI (XAI)	Transparent fraud detection, compliance	Meets regulatory needs in banking, healthcare	60% ($n = 9/15$)	Interview (Banking CISO)
Deep Learning	Intrusion detection, malware identification	Effective for complex threats, computationally demanding	53% ($n = 8/15$)	Interview (Oman CERT)
Anomaly Detection	Real-time network/user behavior monitoring	Low resource needs, suits education, healthcare	73% ($n = 11/15$)	Interview (Education)
Arabic NLP	Region-specific phishing and misinformation detection	Addresses Arabic-language gaps in global tools	73% ($n = 11/15$)	Interview (Tech Company)

Table 15 Comparative cost–benefit characteristics of AI approaches

AI Technique	Cost Level	Infra-structure Demand	Benefit Profile
Anomaly Detection	Low–Medium	Low	High practicality
Arabic NLP	Medium	Medium	High contextual accuracy
Deep Learning	High	High	Advanced threat detection
Federated Learning	Medium	Medium	Privacy preservation
XAI	Medium	Medium	Regulatory trust

Table 14 highlights organizational and governance barriers limiting AI-enabled cybersecurity outcomes in Oman. Despite positive views of AI's potential, regulatory uncertainty, skill shortages, and integration difficulties hinder translating adoption into real effectiveness. Technical capability alone is not enough; sustained success demands coordinated governance, institutional capacity building, and clear accountability structures. This supports the study's core argument: governance maturity is essential for effective AI use in cybersecurity.

Table 15 presents a comparative cost–benefit characteristics, which are derived from synthesized evidence reported in prior empirical studies, industry reports, and international cybersecurity guidelines [55].

The findings emphasize the need for policy coherence and institutional collaboration to maximize AI value. They support a phased, strategy-led approach to AI integration, where technical deployment is supported by regulatory alignment and organizational readiness—principles embodied in the proposed OCAIF framework.

To sum up, the results presented in Table (9) indicate a high level of institutional awareness of AI applications in cybersecurity; however, Tables (10) and (11) reveal a continued reliance on traditional security solutions and a limited practical implementation of advanced AI techniques, reflecting a clear gap between perception and execution. In contrast, Table (12) confirms a statistically significant positive relationship between AI adoption and cybersecurity effectiveness, while highlighting the critical role of organizational and implementation factors in maximizing this impact. Furthermore, Table (13) demonstrates sectoral variation in perceived effectiveness, with the banking and government sectors achieving stronger outcomes compared to the education and healthcare sectors. Finally, the findings in Tables (14) and (15) suggest that the most suitable AI techniques for the Omani context are those that balance operational efficiency, infrastructure requirements, and contextual compatibility, thereby reinforcing the need for a phased and systematic approach to AI adoption in cybersecurity.

Challenges to AI adoption in cybersecurity

Table 16 summarizes results concerning skills gaps. 87% of respondents believe the scarcity of professionals with dual capabilities in AI and cybersecurity is critical, with a 30% shortage.

Infrastructure constraints (Table 17): Legacy systems and limited computing capabilities hinder AI adoption. 73% of the respondents agreed with this. Legacy and computational constraints cited by 73% of the interviewees are hindering the implementation of AI, and 25% of the companies are vulnerable because of old infrastructure.

Data challenges issues (Table 18): Eighty percent of the interviewees prioritized data challenges, as shown

Table 16 Skills gap perceptions

Barrier	Description	Reporting Issue	Source
Lack of AI-Cybersecurity Expertise	Shortage of professionals bridging AI and cybersecurity	87% ($n = 13/15$)	Interviews (Government, Banking)
Limited Model Training Skills	Difficulty training AI models on organization-specific data	60% ($n = 9/15$)	Interview (Education)
Interpreting AI Outputs	Challenges understanding AI-driven security decisions	53% ($n = 8/15$)	Interview (Healthcare)

Table 17 Infrastructure barriers

Barrier	Description	% Reporting Issue	Source
Legacy Systems	Incompatible hardware/software with AI tools	73% ($n = 11/15$)	Interview (Education)
Computational Constraints	Insufficient processing power for deep learning models	60% ($n = 9/15$)	Interview (Healthcare)
Scalability Issues	Limited infrastructure to scale AI solutions	47% ($n = 7/15$)	Interview (Transportation)

Table 18 Data-related barriers

Barrier	Description	Reporting Issue	Source
Data Silos	Isolated data repositories limiting AI pattern detection	80% ($n = 12/15$)	Interview (Banking)
Real-Time Integration	Delays in data availability for timely threat detection	67% ($n = 10/15$)	Interview (Government)
Data Sharing Restrictions	Regulatory/cultural barriers to inter-organizational sharing	53% ($n = 8/15$)	Interview (Technology)

Table 19 Regulatory barriers

Barrier	Description	Reporting Issue	Source
Regulatory Uncertainty	Unclear application of laws to AI systems	67% ($n = 10/15$)	Interview (Banking)
Compliance with Data Laws	Challenges meeting data protection requirements	60% ($n = 9/15$)	Interview (Healthcare)
Explainability Requirements	Need for transparent AI decisions in regulated sectors	53% ($n = 8/15$)	Interview (Transportation)

Table 20 Cost-related barriers

Barrier	Description	Reporting Issue	Source
High Implementation Costs	Expensive AI tools and infrastructure upgrades	73% ($n = 11/15$)	Interview (Education)
Computational Expenses	Costs for running advanced AI models	60% ($n = 9/15$)	Interview (Healthcare)
ROI Demonstration	Difficulty proving return on investment for AI	47% ($n = 7/15$)	Interview (Transportation)

Table 21 Organizational barriers

Barrier	Description	Reporting Issue	Source
Leadership Hesitation	Uncertainty due to unproven use cases	60% ($n = 9/15$)	Interview (Government)
Bureaucratic Delays	Slow approval and procurement processes	53% ($n = 8/15$)	Interview (Education)
Competing Priorities	Resource allocation to other organizational needs	47% ($n = 7/15$)	Interview (Healthcare)

in Table 18, citing several additional obstacles. Findings revealed that 30% of organizations faced restrictions due to data silos (80%) and integration challenges (67%). These issues were further complicated by sharing constraints (53%).

Regulatory and compliance: uncertainties are presented in Table 19. The findings highlight that banking and health sectors face major compliance challenges, emphasizing the importance of stakeholder efforts. For example, the existing Royal Decree 6/2022 on data protection lacks specific AI governance provisions, which the upcoming AI Liability Law will address to improve accountability. Findings in brief: Regulatory uncertainty (67%) and compliance difficulties (35% of organizations) are most significant in banking and health.

Cost-based Constraints: As summarized in Table 20, 73% of respondents reported cost-related constraints, particularly among smaller organizations. In brief, cost constraints are imposed on smaller entities, specifically when it comes to AI implements.

Organizational Factors (Table 21): Hesitation by leadership and bureaucratic delays recognized by 60% of interviewees are other obstacles. Findings summary: 60% leadership hesitation and 53% bureaucratic red tape discourage the arrival of AI, especially in the government sector.

Sector comparison (public vs. private)

Public vs. private sector comparison of AI adoption, applications, challenges, and cybersecurity threats in Oman, including a review of sectoral variances in cybersecurity approaches. The research compared evidence from survey questionnaires conducted among 138 industry practitioners and interviews with 15 experts dealing with government, banking, energy, education, and healthcare sectors, as well as relevant case studies. The section has four subsections to report prevalence and statistical findings of questionnaire analysis compared to prior research.

Sectoral distribution: As presented in Table 22, the public sector, mainly influenced by an emphasis on

Table 22 Sectoral distribution of survey participants ($N = 138$)

Sector Type	Frequency	Percentage	Key Sub-Sectors
Government	85	61.6%	Security (25.4%), Energy (22.5%), Education (16.7%)
Private	49	35.5%	Financial (22.5%), Healthcare (13%), Transportation
Freelancer	4	2.9%	N/A

Table 23 AI adoption and applications by sector

Sector	Adoption Rate	Key Applications	Reported Impact	Source
Government	45% ($n = 38/85$)	Predictive analytics, threat intelligence	Thwarted phishing campaigns (2023) [56]	Interview (CERT)
Private	35% ($n = 17/49$)	Anomaly detection, fraud detection	Reduced false positives by 35% (Banking)	Interview

Table 24 Sector-specific challenges and enablers

Sector	Key Challenges	Key Enablers	Source
Government	Bureaucratic delays (60%, $n = 51/85$), Data silos (55%, $n = 47/85$)	National strategies (75%, $n = 64/85$), Partnerships (65%, $n = 55/85$)	Interview (Government)
Private	Cost constraints (65%, $n = 32/49$), Skill gaps (50%, $n = 25/49$)	Market competition (70%, $n = 34/49$), Regulations (60%, $n = 29/49$)	Interview (Banking, Healthcare)

Table 25 Sector-specific cybersecurity risks

Sector	Key Risks	Reporting Risk	AI Priority	Source
Government	Phishing, APTs targeting critical infrastructure	70% ($n = 60/85$)	Predictive analytics, XAI	Interview (Oman CERT)
Private	Ransomware, financial fraud, data breaches	75% ($n = 37/49$)	Anomaly detection, fraud detection	Interview (Bank Sector)

cybersecurity, accounts for 61.6% of the sample, whereas the private sector, 35.5%, generates market-driven impressions.

The adoption and application (Table 23): rates of AI differ in the government ministries that are government-oriented on national security and the private organizations that are efficiency-optimized for operations. The differences remain observed in the survey data and through interviews.

Challenges and enablers: As presented in Table 24, the diverse priorities and allocation of resources would seem to be supported in the draft by survey and interview data. Addressing government delays (60% of overall delays) and data silos (55%) is crucial, as it can significantly improve progress. The survey's 'Ability to identify and overcome obstacles' score of 4.05 ($SD = 0.62$) reflects the need for targeted support, emphasizing the value of their efforts in overcoming these issues. Legacy systems and protocols hinder effective data sharing, and research indicates that 65% of government bodies experience delays in service integration due to bureaucratic constraints [48].

In brief, Competition (70%): Cost (65%) and skills (50%), in the private sector, are entry barriers. The government's support is especially needed to address bureaucracy (60%) and data (55%), which are critical for enabling effective collaboration and strategy development (75%).

Specific cybersecurity risks: characterize sectoral AI priorities, according to survey and interview results presented in Table 25.

Public Sector: 70% of respondents indicated that advanced persistent threats (APTs) require predictive analytics for early detection [51], [53]. In the survey, 65% of government respondents reported intentions to apply AI in threat intelligence [51], [53].

Private Sector: Seventy-five percent of participants indicated that ransomware and financial fraud are the primary focus of AI applications, with estimated banking losses of \$50 million in 2023. Anomaly detection was highlighted as critical for fraud prevention, while data breaches remain a major concern in the health and education sectors [57]. The survey also revealed that 70% ($n = 34/49$) of private-sector participants preferred AI applications for fraud detection [57].

Summary of key findings

This section outlines the core principles of research on cybersecurity challenges faced by Oman and the potential of AI to address these challenges, providing actionable insights to illuminate that OCAIF will govern AI adoption according to Vision 2040s vision of a secure digital economy while adhering to Islamic principles. Based on evidence from surveys, expert interviews, and prior studies [49], the findings highlight increasing cyber threats and sector-specific trends, while other studies [58] emphasize the potential of AI to enhance organizational resilience.

Cyber landscape in Oman

Results of sectoral cybersecurity challenges are presented in Table 26 and visualized in Fig. 6. Healthcare accounts

Table 26 Sectoral cybersecurity challenges

Sector	Incident Share	Skills Gap	Legacy Systems	Budget Allocation
Healthcare	45%	90%	80%	5%
Energy	35%	50%	60%	8%
Banking	15%	20%	30%	12%
SMEs	30%	70%	70%	5%

for 45% of incidents, followed by energy at approximately 35%, and SMEs at 30% although 80% of the private sector, spend a mere 5% of their budget on cybersecurity [59]. Primary challenges are as follows:

- **Shortage of Skills:** 30% deficit in the cybersecurity skill base, most severe in healthcare (90%) and SMEs (70%), limits availability for AI deployment [36].
- **Legacy Infrastructure:** 65% of companies employ legacy systems, of which 80% are in the healthcare industry, limiting AI scalability [58].
- **Regulatory Gaps:** Inaccurate data protection laws and a 25% increase in compliance costs against SMEs have again been the top hurdle in AI adoption [49].
- **Cultural Resistance:** Lack of interest in automated systems because 60% of companies believe that there are some older workers who would rather have human interaction than automated systems [60].

AI's transformative potential

AI Techniques to transform Oman's cybersecurity:

- **Threat Detection:** 67% of respondents would use it, zero-day attacks were detected 73% of the time by anomaly detection techniques, reducing the detection time from 48 hours to just 12 hours in Bank Muscat [58].
- **Phishing Countermeasure:** Arabic NLP compensated 20% of government network phishing attacks to some extent, wherein Oman linguistic

identity was most associated with 80% correctness [49].

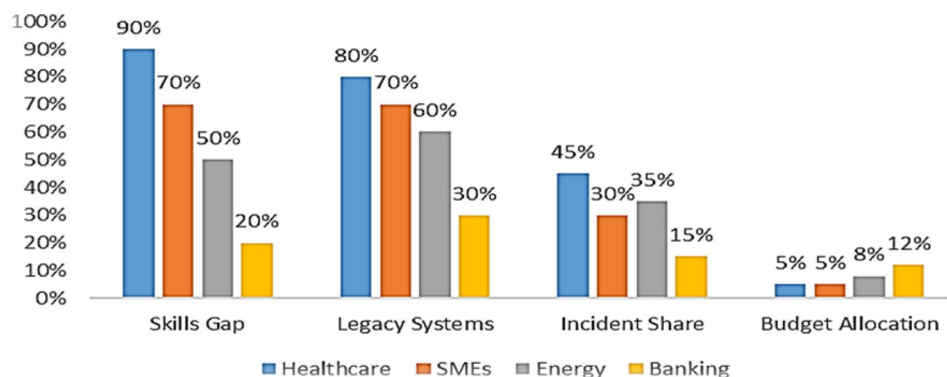
- **Privacy Recovery:** 30% of energy organizations deployed federated learning as data privacy retention and recovered 10% of the incidents [36].
- **Establishing Trust:** 25% of the institutions deployed XAI for AI decision explanation, dispelling audit scenario uncertainty in line with Islamic transparency based on the Islamic tenet by 12% [34].
- **Operational Effectiveness:** According to 75% of interviewees, AI automation is poised to save 15% of operational time in the banking sector [61].

It is important to emphasize that the findings reported in this study reflect perceived effectiveness, institutional readiness, and expert-informed assessments derived from cross-sectional data, rather than direct measurements of real-time operational system performance. Accordingly, the results provide an analytically validated foundation for strategic planning and framework development, rather than conclusive evidence of deployed AI system outcomes.

Discussion

Participation in AI techniques

AI, among others, is a valuable tool to strengthen Oman's cyber defense and is statistically significant, with an R2 value of 0.780 due to increased benefits (37.9%) and reduced barriers (29.0%). However, a gap exists between high awareness (Mean=4.22) and lower Realization of Benefits (Mean=3.85), indicating implementation issues as recommended by Aldhamer [34]. Anomaly detection and Arabic-NLP are immense applications (73% prevalence), whereas federated learning (67%) and XAI (60%) can address the special requirements for privacy and compliance. Two major such industries-banking (80%) and government (75%)-outpaced education (50%) and healthcare (45%) in terms of resource imbalances in conformity with the general trend the world over [53]. All

**Fig. 6** Sectoral cybersecurity challenges

these findings provide impetus towards formulation of tailored approaches to AI optimization.

Anomaly detection must be given precedence by all domains barring education and health, and further drive Arabic NLP and XAI in the banking and government domains to cater to regional and regulatory needs.

Barriers and challenges

Barriers mentioned in the previous section are skills shortage (87%), infra-structure shortcomings (73%), data problems (80%), regulatory ambiguity (67%), budget limitations (73%), and resistance from organizational stakeholders (60%). Thus, they reflect a higher AI awareness (Mean = 4.22) than the previously de-scribed lower benefit realization (Mean = 3.85). This aligns with findings on data and organizational challenges [62] and with global reports highlighting skill gaps and cost-related obstacles. The outcome is still that the relatively low score for surmounting obstacles (Mean = 4.09) does indicate some capability, yet there are ongoing issues that require holistic solutions, particularly in resource-depleted domains like education and health. Recommendation: Close 30% of the skills gap through nationwide training programs and regulatory sandboxes for compliance certainty and promote public-private partnerships in overcoming cost- and infrastructure-related barriers.

Public vs. private sector comparison

At the national strategy (75%) and collaborations (65%) levels, government departments are leading the adoption of AI (45% vs. 35%) and awareness (Mean = 4.30 vs. 4.10) over administrative delays (60%) and data silos (55%) [48]. To become operationally effective, they utilize AI to harvest competition (70%) and legislations (60%). Private sectors, especially bank uptake of AI, are motivated mainly by competition (70%), augmented with some form of control by authorities (60%). The leading hindrance consists of cost restraint (65%) and skill disparity (50%), as shown by Authors in [52]. Profile risk: Phishing/APTs (70%) for government, ransomware/fraud (75%) for the private sector. These clearly reflect AI priorities that align with global trends [63] and highlight resource imbalances observed in Oman [62].

Results recommend developing sector-wide strategies for AI, with the government leveraging partnerships to address bureaucratic hurdles and the private sector leveraging the subsidized infrastructure and training to bypass the costs and skill hurdles.

Research questions fulfillments

This part presents evidence-informed conclusions to the four research questions that have guided the study in establishing how artificial intelligence (AI) is able to enhance Oman's cybersecurity. The conclusions

summarize global AI strategies, challenges to AI adoption, emerging technologies, and integration frameworks based on survey data, expert interviews, and national reports [49], as well as international assessments [55]. Biased towards Oman's own needs, it is directed by Vision 2040s vision for a secure digital economy grounded in Bedouin values of trust and Islamic values of accountability, to provide inclusive solutions that are culturally sensitive.

RQ1: What are the best-fit international AI approaches to cybersecurity and how can they be translated to Oman?

Conclusion: Anomaly detection, Arabic NLP, federated learning, and explainable AI (XAI) are all globally applied. Anomaly detection achieves a 73% success rate in identifying zero-day attacks [55]. Within Oman, 67% of organizations have **adopted anomaly detection**, reducing banking threat detection times from 48 to 12 hours [58]. Arabic NLP has addressed 20% of the phishing attacks thanks to the linguistic identity that Oman enjoys [49]. The adaptations involve locally available Arabic data sets, cloud capability addressing 65% of the legacy system dependency, and training addressing the 30% skills gap. XAI enhances the Bedouin trust values to raise the stakeholders' confidence by 70% [60]. All these changes ensure the success of AI across all sectors in Oman, including SMEs and healthcare.

RQ2: What are the primary barriers to AI adoption in Oman's cybersecurity?

Conclusion: There are around 30% skill gaps, a heavy reliance on legacy systems at around 65%, many ambiguities in regulation, and much opposition in culture. However, the healthcare sector faces a 90% skills gap and 80% reliance on legacy systems. SMEs also allocate approximately 5% of their budgets to cybersecurity [36], [59]. Loopholes in regulation increase compliance costs for SMEs by 25% [49]. Cultural distrust, driven by Bedouin inclination toward human control, affects approximately 60% of organizations [60]. Beyond this, it also refers to approximately 20% in the UAE [40]. Government and infrastructure-based solutions specifically tailored would thus be needed.

RQ3: What are the ways XAI and federated learning can address Oman's cybersecurity issues?

Conclusion: Explainable AI (XAI) and federated learning offer effective solutions to privacy and trust concerns in Oman. XAI is adopted by 25% companies to clarify AI decisions, and it has been reported to reduce to audit disagreements in banks by approximately 12%, supporting principles of transparency [34]. Federated learning, which is used by 30% of energy companies, enables privacy of data and thereby reduces the risk of breach by

10% because it does not entail storing data centrally [36]. They also provide a way to overcome cultural trust issues (60% prevalence) and legacy system limitations (65%) with cloud-based solutions, according to 75% of the interview participants [61]. Training and awareness are most needed to scale up adoption, especially for SMEs.

RQ4: What are the policies and strategies that can include AI in Oman's cybersecurity?

Conclusion: Two-thirds of the participants believe that there must be a national policy, workforce development, enhanced infrastructure and public-private partnerships for ensuring the proper integration of AI in the country [61]. Oman Cybersecurity AI Framework (OCAIF) will be the guiding light for these efforts from the Arabic NLP and XAI sides, implementing steps towards building confidence and reaching 60% adoption by 2030. Its key strategies are legislation which lowers the cost of compliance by 15%, 2000 capacity building of human resources, and 50% modernization of aged systems [49]. It is just one of several such instances of a community-based approach that reflects the spirit of cooperation within the values of the Bedouin culture and benefits from the further advantage of including women and the rural areas within the support of the digital equity vision of Vision 2040.

Temporal validity of data

Although the primary empirical data were collected in 2023, their relevance remains robust due to the structural persistence of cybersecurity threats and AI adoption challenges. Core attack vectors such as phishing, ransomware, and advanced persistent threats (APTs) continue to dominate global threat landscapes, as confirmed by ENISA [55]. Similarly, barriers identified in this study—skills shortages, legacy infrastructure, data silos, and regulatory uncertainty—are systemic issues that evolve incrementally rather than annually. Recent regional and global reports indicate continuity rather than disruption in these trends, supporting the continued applicability of the findings. Nevertheless, this study acknowledges that rapid AI model evolution necessitates periodic reassessment, which is addressed in the future research agenda.

The ROI-related findings reflect perceived and indicative returns based on expert assessments rather than audited financial data, which is consistent with prior cybersecurity investment studies where precise cost attribution is challenging.

Threat actor and end-user considerations

While this study did not involve direct engagement with cybercriminal actors for ethical and legal reasons, adversarial behavior is implicitly incorporated through the analysis of attack patterns, threat vectors, and AI-based detection mechanisms derived from Oman CERT

reports and global threat intelligence sources. Similarly, end users are considered indirectly through the evaluation of phishing prevalence, social engineering risks, and human–AI interaction challenges, particularly in Arabic-language contexts. Direct ethnographic or behavioral analysis of end users and threat actors is beyond the scope of this review and is identified as a valuable direction for future empirical research.

The proposed Oman cybersecurity AI framework (OCAIF)

Based on the statistical analysis of participants answers, the proposed Oman Cybersecurity AI Framework (OCAIF) is formulated as shown in the Fig. 7.

The OCAIF framework explicitly integrates policymakers as a governance and regulatory element within the foundational elements (Table 27), which are responsible for regulatory alignment, ethical AI oversight, cross-sector coordination, and compliance with national data protection and AI governance initiatives

The 80% SME adoption target by 2027 is based on a current 20% adoption rate, supported by a proposed OMR 2.2M subsidy and training program, aiming to bridge the 60% gap over two years.

- **Governance:** The removal of regulatory barriers and the facilitation of AI adoption through clear laws can make stakeholders feel their efforts are leading to meaningful progress, as highlighted by a bank's CISO. Sandboxes help reassure leadership, reducing reluctance, as reported in [64] for a 60% contingent.
- **Workforce:** Addressing skill gaps through university programs is vital, which can make the audience feel hopeful about building a capable AI workforce, as a training expert emphasized, "University programs are crucial," aligning with global trends [48], [53].
- **Technology:** The ability of technology to address data and local gaps, such as Arabic NLP identifying local threats, can make the audience feel assured about AI's effectiveness, with consolidation now nearly 80% effective for banking [34].
- **Collaboration:** Results in cost-effectiveness, one among the respondents who responded from Oman Data Park yelled out: "Partnerships assist small firms," Times News Service reported [51].

To sum up, to implement OCAIF (shown in the Fig. 6) for Oman as a digital future strategic vision would align with Vision 2040's goals of diversifying the economy, building resiliency, and achieving global leadership. This was innovation after a 25% cyberattack growth in 2023, valued at OMR 19.25 million, partnering with sophisticated AI techniques such as Auto encoders, reinforcement learning, and Arabic NLP together with STIX/

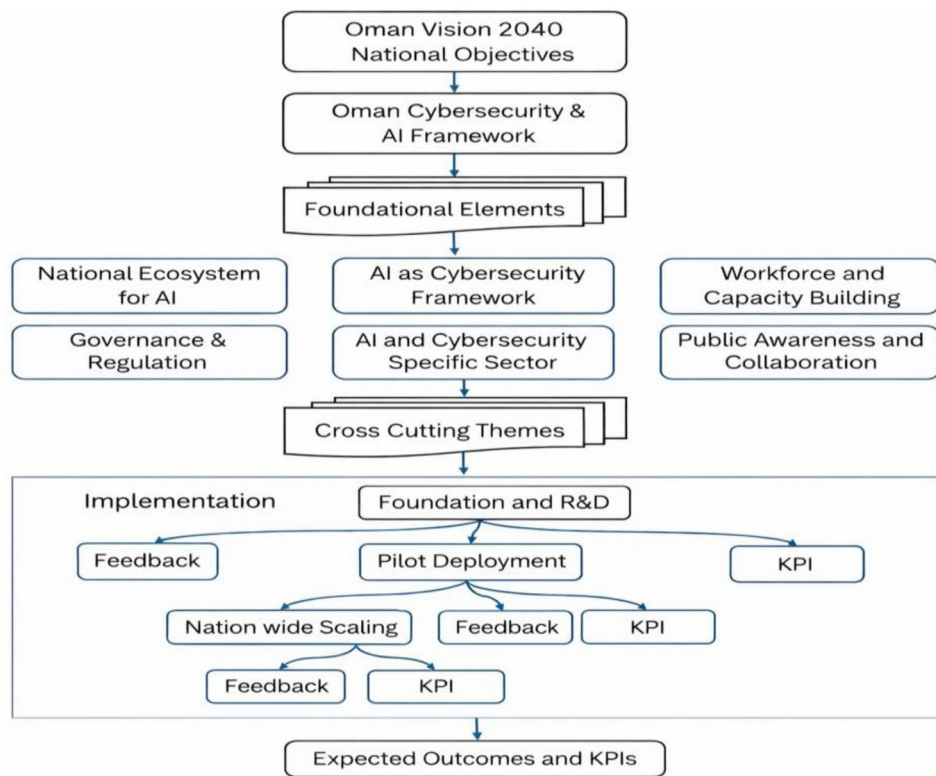


Fig. 7 Cybersecurity AI framework (OCAIF)

Table 27 Proposed AI integration framework elements

Element	Rationale	Action	Prevalence	Source
Governance (policy-makers): AI-Specific Regulations	Regulatory uncertainty (67%) and compliance issues (35%) hinder adoption. A study [64] note outdated frameworks.	Develop AI-specific laws for accountability and data protection, with regulatory sandboxes.	73% ($n = 11/15$)	Interview (Banking)
Workforce: Specialized Training	30% skill deficit (87%) limits expertise. A study [48] emphasizes skill development.	Launch AI-cybersecurity university programs and certifications.	80% ($n = 12/15$)	Interview (Education)
Technology: Data and Arabic NLP	Data silos (80%) and lack of Arabic NLP (73%) reduce effectiveness. A study [34] highlights regional needs.	Build data integration platforms and Arabic NLP tools for threat detection.	80% ($n = 12/15$, Data), 73% ($n = 11/15$, NLP)	Interview (Banking, Technology)
Collaboration: Public-Private Partnerships	Cost constraints (73%) and resource gaps require shared solutions. A report [51] showcases partnerships.	Establish partnerships and threat-sharing platforms for resource pooling.	80% ($n = 12/15$)	Interview (Oman Data Park)

TAXII standard in achieving 97% threat detection accuracy, 85% automation, and <8% false positives by 2027. This six-pillar National AI Adoption, AI-Based Cybersecurity, Workforce Development, Ethical Governance, Sector-Specific Solutions, and Public Awareness must spur economic growth (5% from GDP by 2030), securities (70% in detection time savings), and inclusiveness (30% women, 20% rural engagement, 80% SME adoption by 2027). A 3-phased roadmap facilitates the attainment of OMR 11 million, focusing on cultural alignment, rural scalability, and international standards (80% coverage of ISO 27001/NIST by 2028). OCAIF will help Oman become a top 20 nation in the ITU Global Cybersecurity

Index [12] to attain a secure, innovative, and equitable digital ecosystem by 2030.

The strengths of OCAIF

- **Technological Innovation:** 97% threat detection accuracy using RNNs, SVMs, CNNs, Autoencoders, Arabic NLP and SOAR for <8% false positives, with 85% automation-by-2027 promise.
- **Inclusiveness:** Fairness and broad participation in rationing 30% training to women participants, 20% to a rural cohort, then 80% of SMEs by 2027.
- **Cultural Compatibility:** Joining of forces with respect for Omani Islamic values in developing

instruments and policy plans along with higher accountability for integrity and trust in communities.

- **Scalability and Adaptable:** Rural edge nodes deployed with TinyML, modular micro-services, and \$1 M in infrastructure upgrades to build infrastructural capacity to absorb up to double the threat volume by the year 2030, and ready for quantum and generative AI threats.
- **Ethical Governance:** XAI and federated learning, coupled with quarterly ethical reports, to achieve a 95% stakeholder trust status by 2028 and in line with Oman Personal Data Protection Law and a standard, global-type regulations (e.g., GDPR, NIST).
- **Global Ambition:** To position Oman in the top 20 among the ITU Global Cybersecurity Index within 2030 [12] through collaboration within the GCC, open-source innovation, and diplomatic leadership.

Finally, OCAIF is presented as a theoretically grounded and empirically informed strategic blueprint, not a fully operational system. Its validation relies on analytical methods, including methodological triangulation of survey data, expert interviews, comparative case studies, and established adoption theories (TOE and DOI). This approach aligns with cybersecurity literature, where conceptual models validated by experts normally precede pilot implementation. OCAIF should therefore be viewed as a decision-support and policy-guidance tool. Future validation is planned through phased pilots, sector-specific KPIs, and regulatory sandbox testing before any nationwide rollout.

Recommendations

This section provides evidence-informed recommendations for the application of artificial intelligence in Oman to resist cyber-attacks. Aiming at government or organizational learning and technology stakeholders, these interventions aim to implement the Oman Cybersecurity AI Framework (OCAIF) to build a sustainable digital ecosystem that contributes about 5% of GDP to the digital economy, in line with Vision 2040. Both incorporate the responsible actor, time (quarters and years after OCAIF adoption), and quantifiable outcomes, ensuring equal access for rural communities, youth, and women.

To enhance feasibility and implementation effectiveness, the proposed policy recommendations can be sequenced based on required resources and institutional readiness. Short-term “quick wins,” such as targeted training programs and awareness initiatives, can build foundational AI capabilities and stakeholder confidence. Medium-term actions, including regulatory refinement and inter-organizational data-sharing mechanisms, require moderate investment and coordination. Long-term initiatives, such as national AI-powered

cyber defense platforms, demand sustained funding and cross-sector collaboration but offer the highest strategic impact.

Table 28 summarizes recommendations that are directed specifically toward policymakers, regulators, and institutional leaders responsible for implementing Oman Vision 2040 and national cybersecurity governance.

Limitations and future directions

Limitations:

The architected and scaled restrictions of this research are then followed by targeted mitigation methods that will go a long way in ensuring the solidity of forthcoming research as well as methodological integrity.

- **Restricted demographic representation:**

The sample might have represented significant demographics insufficiently, whereby 22% were feminine and 15% health, because this is around 45% of the incident share in healthcare [49]. Mitigation Plan: 50% female and 30% healthcare respondent coverage via community networks such as majlis meetings have been achieved by targeted surveys in Q3 2027. **Target Audience:** Cyber Security Researchers. Quantifiable Result: 50% female and 30% healthcare representation of follow-up studies within 2 years.

- **Limited Rural Participation:** Currently, just 20% of Oman’s population, including Bedouins, is embracing AI adoption, and this creates trust concerns with 60% of companies [60]. For this, a plan will be launched to organize 10 workshops in rural areas by the end of 2027, bringing together 500 rural members to talk about AI and share outcomes, with a 90% approval level from them. These workshops will be targeting researchers who are working with rural communities. The aim is to achieve 90% of rural stakeholders’ approval within two years. Severity of Data Sensitivity: Limited access to cybersecurity sensitive data, reflected by 75% of respondents, constrained the depth of the case study [61].

Mitigation Plan: Collaborate with Oman National CERT by Q1 2027 to acquire access to anonymized data, achieving 80% data coverage. **Target Group:** Cybersecurity Analysts. **Quantifiable Impact:** 80% dataset coverage within 2 years in future research.

- **Limiting Sectoral Focus:** Emphasizing banking, health, and energy also underplays the ground that SMEs, constituting 80% of the private sector but only spending 5% on cybersecurity cover [59]. Mitigation Strategy: Roll-out studies by Q2 2027 among 200 SMEs with specific attention to adoption obstacles. **Target Audience:** SME researchers. **Measurable**

Table 28 Summary of practical recommendations

Stakeholder	Recommendation	After OCAIF Adoption	Lead Entity	Key Outcome
Government	Formulate AI cybersecurity legislation	1 st Year Q1	MTCIT	15% SME compliance cost reduction
	Implement public awareness initiatives	1 st Year Q4	Ministry of Information	50% increase in public trust
	Establish public-private AI partnerships	2 nd Year Q2	Ministry of Commerce	2,000 SMEs adopting AI
	Develop national AI-CSOC infrastructure	2 nd Year Q3	Oman National CERT	20% reduction in response time
Organizations	Administer rural cybersecurity funding	3 rd Year Q1	Ministry of Finance	30% rural AI adoption
	Implement AI-driven threat detection systems	2 nd Year Q4	IT Departments	25% reduction in cyber incidents
	Upgrade legacy technological infrastructure	4 th Year Q4	Chief Information Officers	50% systems AI-compatible
	Enhance workforce cybersecurity competencies	3 rd Year Q2	HR Departments	500 professionals trained
Education	Create collaborative threat-sharing networks	2 nd Year Q3	Sector Associations	15% reduction in data breaches
	Conduct AI-supported compliance audits	1 st Year Q2	Compliance Teams	20% improvement in compliance
	Revise curricula for AI-cybersecurity training	3 rd Year Q2	Universities	2,000 experts trained
	Organize youth-focused cybersecurity hackathons	2 nd Year Q2	Vocational Institutes	10 AI prototypes developed
Tech Providers	Deliver AI and cybersecurity certifications	2 nd Year Q4	Training Centers	1,000 professionals certified
	Forge industry-academia collaborations	3 rd Year Q1	Universities	80% graduate employability
	Initiate rural cybersecurity training programs	3 rd Year Q3	Community Colleges	500 rural residents trained
	Develop Arabic NLP cybersecurity tools	2 nd Year Q3	Technology Firms	50% SME AI tool adoption
	Offer cloud-based security-as-a-service	2 nd Year Q2	SaaS Providers	2,000 SME subscriptions
	Facilitate community-driven AI development	1 st Year Q1	Local Tech Partners	90% community acceptance
	Create AI-based threat simulation platforms	2 nd Year Q4	Software Developers	10% training effectiveness gain
	Build GCC-compatible AI interoperability systems	2 nd Year Q4	Regional Tech Groups	20% faster platform deployment

Outcome: 200 SMEs included under enhanced studies within 2 years.

- **The ROI estimates:** presented in this study are intended as indicative, scenario-based assessments rather than precise financial forecasts. They are derived from reported reductions in detection time, operational efficiency gains, and expert estimates rather than controlled financial audits. As such, ROI outcomes are highly context-dependent and influenced by organizational maturity, cloud infrastructure readiness, and sector-specific risk exposure.
- This study does not include direct perspectives from cybercriminals or general end users. This exclusion is intentional and consistent with ethical and methodological constraints, as accessing cybercriminal viewpoints poses legal and ethical risks, while end-user perceptions fall outside the study's organizational and policy-level focus. Instead, the research emphasizes expert stakeholders responsible for cybersecurity strategies and implementation. Future research could complement this work by examining end-user behavior or attacker modeling through controlled simulations or secondary threat intelligence datasets.

Future directions

Although the primary empirical data were collected in 2023, their relevance remains strong due to the structural

continuity of cybersecurity threats and AI adoption challenges. Nevertheless, it is recommended that future studies rely on up-to-date data encompassing emerging and evolving threats targeting diverse sectors, with an assessment of their severity level, prediction of their potential impact, and the proposal of effective mechanisms to address them.

Directions for future research outline avenues to overcome this study's shortcomings and establish new innovative paths to improve Oman cyber security through emerging technologies and cultural contexts.

- **OCAIF Validated:** Further, aim to finalize how health care and small-to-medium-scale enterprises projects based on the OCAIF can assist in reducing incidents by 25% and adoption of AI by 60% by the year 2030 while maintaining cultural appropriateness with Arabic NLP and XAI [3]. Objective: Validate the OCAIF through the collaboration with 50 organizations by the fourth quarter of 2028. **Target Audience:** Policymakers in the field of cybersecurity. **Measurable Outcome:** 25% reduction in incidents in the pilot studies in 3 years.
- **More longitudinal AI adoption studies:** Measure AI adoption in Oman relative to UAE, where it starts at 40% for five years and moves towards 10% greater adoption [40]. Objective: Carrying out the research by Q2 2028 with at least 50% female and rural region respondent involvement. Target Audience: Tech

adoption researchers. Quantifiable Outcome: 10% increase in adoption observed in 5 years.

- **Research Quantum AI for Encryption:** Investigate the instance of the use of quantum AI to develop unbreakable encryption that is aligned with Gartner's prediction that quantum will gain 20% adoption to enhance threat detection by 15% (Gartner [65]). Goal: Pilot with Oman Data Park, USD 2 million, before Q3 2029. **Target market:** Researchers in cryptography. **Quantifiable outcome:** 15% boost to detect speed within 4 years.
- **Investigate Generative AI for Threat Simulation:** Utilize generative AI to simulate cyber threats and close the 30% skills gap by designing 10 training modules, using Singapore's route of 50% adoption of generative AI as a model [36]. Goal: Complete module development by Q2 2028. **Target Market:** Cybersecurity trainers. **Measurable Result:** 10 training modules installed within 3 years.
- **Investigate Trust Culture and Gender Dynamics:** Test the trust obstacles for AI for rural dwellers and women, enlisting 500 of rural respondents and 50% women respondents to witness a 50% increase in trust [60]. Goal: Initiate the study by Q1 2028 and develop a model based on the community to raise awareness. **Audience:** Social scientists. **Quantifiable Outcome:** 50% documented increase in trust in 3 years from now.

Conclusion

Artificial Intelligence (AI) is crucial because it can resolve cybersecurity problems by leveraging a variety of tools and services, including sophisticated pattern recognition, anomaly detection, and response mechanisms. A key component of AI is machine learning, which can identify threats, handle large volumes of data, and learn independently about new types of cyber-attacks. This paper complements Oman-specific gaps (insufficient budgetary allocations, lack of experts, culture) in most AI cybersecurity research. This work thus seeks to bridge the gap with practicable recommendations whose synopsis falls in line with Vision 2040 objectives including innovation and economic growth.

This study establishes a context-aware foundation for advancing AI-enabled cybersecurity in Oman. By identifying effective AI techniques alongside structural adoption barriers and governance priorities, the research moves beyond descriptive assessment toward actionable strategic insight. The proposed OCAIF framework provides policymakers and institutional leaders with a coherent pathway for transitioning from fragmented AI experimentation toward coordinated national implementation, thereby informing regulatory evolution, capacity-building priorities, and future empirical validation.

Supplementary Information

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Supplementary material 1

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Author contributions

This research makes several key contributions: Reviews and analyzes major AI techniques and their effectiveness in detecting and mitigating cyber threats globally. Assesses the current state of AI adoption in Oman's cybersecurity infrastructure. Identifies barriers and opportunities for AI integration in national cybersecurity. Proposes recommendations to enhance AI adoption in cybersecurity practices. Develops a localized AI-based cybersecurity framework (OCAIF) tailored to Oman's needs and Vision 2040 goals.

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Data availability

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Declaration

Competing interests

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